Worksheet 51 · Arithmetic Sequences and Series IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Find the nth term of an arithmetic sequence • Find the sum of an arithmetic series • Solve problems using arithmetic series formulae		
Grade: / 62	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Key notes

Notation → a = first term d = difference n = position

Summation → $S_n = \frac{n}{2}[2a+(n-1)d]$

N^{th} term → $a_n = a+(n-1)d$

Questions *Arithmetic Sequences and Series*

Question 1 **(4 marks)**

Work out the sum of the multiples of 3 between 1 and 1000

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Question 2 **(5 marks)**

The 25th term of an arithmetic series is 44.5
The sum of the first 30 terms of this arithmetic series is 765

Find the 16th term of the arithmetic series.
Show your working clearly.

Question 3 **(4 marks)**

The 10th term of an arithmetic series, S, is 66
The sum of the first 20 terms of S is 1290

Find the 5th term of S.
Show your working clearly.

Question 4 **(4 marks)**

$(2x + 23)$, $(8x + 2)$ and $(20x - 52)$ are three consecutive terms of an arithmetic sequence.

Prove that the common difference of the sequence is 12

Question 5 **(5 marks)**

The sum of the first 48 terms of an arithmetic series is 4 times the sum of the first 36 terms of the same series.

Find the sum of the first 30 terms of this series.

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Question 6 **(4 marks)**

An arithmetic series has first term 1 and common difference 4

Find the sum of all terms of the series from the 41st term to the 100th term inclusive.

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Question 7 **(4 marks)**

An arithmetic series has first term a and common difference d.

The sum of the first $2n$ terms of the series is four times the sum of the first n terms of the series.

Find an expression for a in terms of d.
Show your working clearly.

$a =$ .....

Question 8 (5 marks)

The sum of the first 10 terms of an arithmetic series is 4 times the sum of the first 5 terms of the same series.

The 8th term of this series is 45

Find the first term of this series.
Show clear algebraic working.

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Question 9 (5 marks)

Here are the first four terms of an arithmetic series.

$$k \quad \frac{3k}{4} \quad \frac{k}{2} \quad \frac{k}{4}$$

Given that the 15th term of the series is $(90 + 2k)$,

calculate the sum of the first 30 terms of the series.

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Question 10 (5 marks)

The sum of the first N terms of an arithmetic series, S, is 292
The 2nd term of S is 8.5
The 5th term of S is 13

Find the value of N.
Show clear algebraic working.

$N =$ .....

Question 11 (5 marks)

An arithmetic series has first term a and common difference d, where d is a prime number.

The sum of the first n terms of the series is S_n and

$S_m = 39$

$S_{2m} = 320$

Find the value of d and the value of m
Show clear algebraic working.

$d =$ .....

$m =$ .....

Question 12 (6 marks)

The nth term of an arithmetic series is u_n where $u_n > 0$ for all n
The sum to n terms of the series is S_n

Given that $u_4 = 6$ and that $S_{11} = (u_6)^2 + 18$

find the value of u_{20}

Question 13 (6 marks)

A polygon has n sides, where $n > 5$

When arranged in order of size, starting with the largest number, the sizes of the interior angles of the polygon, in degrees, are the terms of an arithmetic sequence.

Here are the first five terms of this sequence.

177 175 173 171 169

Find the value of n
Show clear algebraic working.

$n =$ .....